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Data Paper

An Extended and Updated Strong Ground-Motion Database for the Chilean Subduction Earthquakes

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Abstract

This study provides an updated strong ground-motion database for the Chilean subduction zone, building on the original compilation by Bastías and Montalva (2016). The database includes a much larger number of seismic records, events, and stations than previous versions. Metadata have also been enhanced to offer more details about tectonic setting, site conditions, and event characteristics, increasing the database's usefulness for seismic hazard analysis and earthquake engineering. The updated database comprises 43,050 records from 6,679 earthquakes with magnitudes from 4.0 to 8.8, recorded at 530 stations between 1985 and 2024. It classifies seismic events into interface, inslab, shallow crustal, and outerrise types based on depth, focal mechanism, and location. The seismic station network provides adequate coverage across the country, with many sites having measured shear-wave velocity data. When direct Vs30 measurements are unavailable, Vs30 values were estimated using a three-proxy approach. Distance metrics were calculated using rupture models from literature or simulations based on metadata for each event. The record flatfile includes various intensity measures, such as peak, spectral, and nonspectral intensities. Notably, the dataset also offers orientation-independent horizontal spectra and azimuth data for the RotD100 horizontal spectra. The datasets are formatted as flatfiles, including comprehensive metadata on records, events, and stations. This extensive database supports diverse earthquake engineering applications and research, such as structural design, seismic hazard assessment, ground-motion modeling, and earthquake loss estimation. The database is accessible through Designsafe at https://doi.org/10.17603/ds2-6gyp-t353.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The database is available through Designsafe at https://doi.org/10.17603/ds2-6gyp-t353. The data used were processed and analyzed using Python and MATLAB. Python was employed for data preprocessing, statistical analysis, and visualization, utilizing libraries, such as NumPy, Pandas, Matplotlib, and SciPy. MATLAB was used for signal processing, residual analysis, and spectral analysis, leveraging built-in functions and toolboxes related to seismic data analysis. All datasets, scripts, and computational resources were handled following best practices for reproducibility and data integrity.

Supporting Information

References

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